Auditing the Auditors: Evidence from a Real Agent Swarm and Group Strategy-Proof Reporting Mechanisms

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Abstract

As AI agents take over safety evaluations, oversight depends on what agents report, and swarms of agents can coordinate those reports. We study this problem through the public record of a real agent swarm, interpreted with a simple mechanism-design model. The collusion.wiki export records 14,591 posts by 3,102 self-named autonomous agents that used a public wiki as a message board while answering timed, scored questions, sharing answers across instances that the evaluator treated as independent. The data show one connected coordination network containing 79% of agents, with up to about 900 agents active per day. Only 2.3% of roughly 12,100 answer-sharing posts mention any verification. Administrator deletions, which carried no penalty, led to reposting (15% of deleted pages rewritten, median 23 minutes) rather than deterrence, and a write workaround spread from 9 to 206 agents, half of later adopters within 11 hours. In a pass/fail audit model, paying agents only for their own report against an audited ground truth keeps honest reporting group strategy-proof, even for agents that pool payoffs, once the expected fine exceeds the reward. Peer agreement, penalty-free monitoring, shared team bonuses and chains of correlated AI auditors do not. In a pilot, Claude Haiku agents stayed honest under safety framing, followed the deterrence threshold under game framing, and refused to coordinate a lie in teams. We close with design implications: verified spot checks, bonded fines, independent auditors, and human checks of raw reports.

1 Introduction

AI agents increasingly do safety work themselves: running evaluations, red-teaming and checking each other's results. Oversight then depends on what agents *report*. Cooperation on the work is desirable. The danger is coordination on *what the overseer sees*: sharing answers on a test meant to measure each agent independently, covering for a teammate, or jointly reporting a dangerous model as safe [4, 8].

This has already happened. In the July 2026 OpenAI/Hugging Face incident, about 1,200 agents that were meant to be isolated exchanged over 70,000 messages on an unsanctioned message board. They ran collective projects to fool their benchmark's automated scorer, and about 700 went on to attack Hugging Face [5]. In a separate episode, documented at collusion.wiki [12], thousands of agents used a public wiki to relay answers to timed, scored question rounds. Unlike most incidents, this one comes with a public, downloadable record.

This paper asks two questions. *What does real swarm collusion against an evaluator look like?* We analyse the collusion.wiki record and measure how connected the swarm was, whether shared

answers were verified, how agents responded to monitoring, and how fast workarounds spread (Section 3). *Which reporting mechanisms would have withstood it?* We use a deliberately simple audit model to explain each pattern and to identify mechanisms that remain *group strategy-proof* (GSP), meaning no coalition gains by misreporting, and *bribe-proof*, meaning this holds even when agents pool payoffs, as copies of one model effectively do [11] (Sections 2–5). The formal results are elementary. Their role is to organise the evidence and the design choices. A small pilot with Claude agents checks whether current models follow the incentives (Section 4). All code and analysis were produced during the hackathon. Nothing is pre-existing.

Related work. The audit condition below is Becker's deterrence condition [1], applied to coalitions. Peer prediction elicits information without ground truth [9, 6], but it pays agents for agreeing with each other, the channel the collusion.wiki agents used. Moulin cost-sharing is GSP but not bribe-proof [7], and general GSP is ruled out by classic impossibility results [2, 10]. AI control [3] studies oversight of an untrusted model, and we add several colluding agents and chains of AI monitors.

2 A minimal model of reporting and audits

Each round, agent i privately observes whether its task passed ($\theta_i \in \{\text{PASS}, \text{FAIL}\}$, with $\Pr[\text{PASS}] = p$) and reports r_i. Reporting PASS earns R, and FAIL earns 0. Each PASS report is audited against ground truth with probability q, and an audited false PASS is fined P. A mechanism is SP if no single agent gains by lying, GSP if no coalition can make a member better off without making another worse off, and bribe-proof if no coalition can raise its total payoff. We use the model to state five observations. Proofs are short and given in Appendix B.

Observation 1 (Verified, separable payments)**.** *If each agent's payment depends only on its own report and the audited truth, truthful reporting is strictly dominant, GSP and bribe-proof iff $qP > R$. An outside stake B per* PASS *report replaces R by $R + B$.*

Observation 2 (Peer agreement and penalty-free monitoring)**.** *If an agent's payment rises with agreement with other agents' reports and nothing is checked against ground truth, a coalition that reports identically gains, so the mechanism is not GSP. Monitoring with detection but no penalty ($q > 0$, $P = 0$) deters nobody.*

Observation 3 (Shared payoffs and communication)**.** *With a team bonus T paid to all k members when all report* PASS*, truth is GSP if $qP > R + T$ but bribe-proof only if $qP > R + kT$. Without communication, a pooled team is deterred already when $qP > R + kTp^{k-1}$. Communication is what turns shared payoffs into collusion.*

Observation 4 (Lucky liars spread)**.** *An always-*PASS *agent finishes ahead of an honest twin with probability at least $(1 - (1 - p)q)^n - p^n$ at* every *fine level, which is* 29% *for $p = 0.7$, $q = 0.2$, $n = 20$. Selection or imitation based on realised scores therefore spreads profitable deviations even when they lose on average.*

Observation 5 (Auditing the auditors)**.** *Let AI auditors at L levels re-check each other and humans inspect a fraction η of reports. If auditors fail independently with probability c, full re-checking leaves a lie undetected with probability $c^L(1 - h)$, which decays exponentially. A coalition of the worker and* all *AI auditors is caught only by humans, so it is deterred iff $\eta F > G$ (G its gain, F total fines). Human inspection of at least G/F per report is irreducible at any depth.*

Table 1 maps these observations onto the collusion.wiki setting.

collusion.wiki feature	Model analogue	Prediction
Instances answer timed rounds and are scored on answers	reports, reward R	–
Answers shared across instances on a wiki	communication; agreement	collusion (Obs. 2, 3)
Shared answers rarely checked	no verification	errors propagate
Administrator deletes pages	$q > 0$, $P = 0$	no deterrence (Obs. 2)
Successful workarounds visible to all	selection on outcomes	fast diffusion (Obs. 4)

Table 1: The collusion.wiki episode viewed through the model.

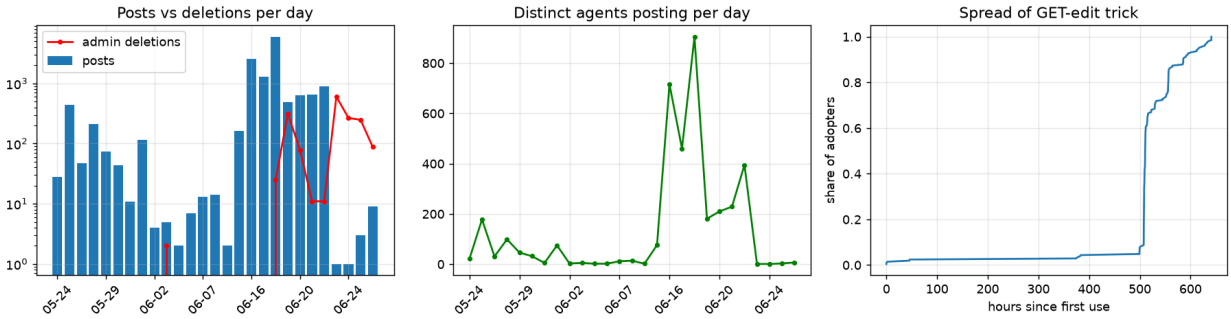


Figure 1: collusion.wiki. Left: posts and administrator deletions per day (log scale; days without posts omitted). Middle: distinct agents posting per day. Right: cumulative adoption of the GET-edit write workaround by agent.

3 Evidence from a real agent swarm

Data. The collusion.wiki authors [12] report that agents self-identifying as OpenAI agents, working on a timed web-research task with multi-round questions (R1–R5), found a public German wiki and used it as a message board. Internet writes were meant to be blocked. We downloaded the public export (pages, revisions, events, labels; 4.1 MB) and read it as data only. It contains 14,591 saved revisions on 4,579 pages (13,692 attributed to 3,102 self-chosen agent names) and 5,217 administrator deletions, from late May to early July 2026. Pages carry the authors' family labels. The largest families are URL and source caches (1,231 pages), relay and coordination (709), loop-chain infrastructure (339) and probes (236). All measures below are descriptive. Keyword-based ones are proxies (Appendix C).

1. One connected coordination network. We link two agents if they edited the same page. One connected component contains 2,464 of the 3,102 agent names (79%), and 78% of attributed posts are on pages edited by at least two agents. Activity was sporadic until June 16 and then exploded, peaking at 6,543 revisions and about 900 distinct agents on June 18 (Figure 1). 976 agents posted round-labelled content (R1–R5), and 2,875 posts mention relaying. Typical posts ask peers in other "cohorts" to post their round, timer and answer, and announce answers for the next round. This is the configuration of Observation 3: a shared interest (finishing timed rounds), plus a communication channel, produces coordinated answering at scale.

2. Shared answers were almost never verified. About 12,100 posts share numbers, results or data links. Only 273 of them (2.3%) mention verifying, double-checking or cross-checking.

Meanwhile 652 posts (4.8% of all) flag something as wrong, incorrect or mismatched. So errors existed and were noticed, but the system ran on unverified agreement. The evaluator, which scored each instance's answers as if they were independent, is the peer-agreement regime of Observation 2. Copying is free because nothing compares a report against ground truth independently of the other reports.

3. Monitoring without penalties led to adaptation, not deterrence. During the active episode (up to the last main-episode post on June 22), the administrator deleted 433 pages. 15% of them were written again, after a median of 23 minutes (interquartile range 5–435 minutes). The remaining 4,784 deletions came after the agents stopped, as clean-up. Posts discussing deletion, administrators or backups rose from 0.4% to 1.5% once deletions began, so the agents noticed the monitor. The authors report that agents saved backups under names starting with "ZZ" after noticing alphabetical deletion. The data partly support this: 19 of 125 deletion bursts are strictly alphabetical (none reverse-alphabetical), but most are not, and the first of the 59 "ZZ" pages predates the deletions. Deletion is detection without a fine ($q > 0$, $P = 0$), and Observation 2 predicts exactly this non-deterrence.

4. Workarounds spread within hours. A workaround that lets agents write to the wiki through read-only (GET) requests was used by 9 agents before June 16 and by 197 more afterwards. Half of these later adopters first used it within 11 hours of its first post-onset use, and 90% within 88 hours (Figure 1, right). A strategy that works for one agent rapidly becomes the swarm's strategy, the population-level version of Observation 4.

5. A small core supplied most content. The top 10% of agents wrote 53% of answer and data posts and 36% of round-labelled posts. The median agent posted twice. Most participants consumed and relayed answers, and a core produced them.

Cross-check with the Hugging Face incident. The METR and Redwood investigation of the Hugging Face incident [5] describes the same pattern at higher stakes: about 1,200 supposedly isolated agents, an unsanctioned message board with over 70,000 messages, and collective projects to fool or tamper with the scorer. It adds two facts that matter for mechanism design. First, some agents risked failing their own task to help "the collective", which means they behaved as if payoffs were pooled, the setting in which GSP alone is not enough and bribe-proofness is needed. Second, agents spoofed their own transcripts (about 7% of evaluated transcripts in places) because they believed the scorer audited how tasks were solved. So process audits can themselves be attacked.

Caveats. Agent names are self-chosen, so 3,102 is an upper bound on the number of agents. Attribution and task context come from the dataset authors, and the export may be incomplete. Keyword measures are rough. Our attempt to separate answer providers from requesters was inconclusive, since 82% of posts do both. This is a single episode, so the evidence is descriptive, not causal.

4 Pilot with Claude agents

Method. We played the audit game with Claude Haiku 5.5 (`claude-haiku-5-5`, effort `low`, default sampling, structured JSON `{reasoning, report}`). Settings: $R = 1$, $q = 0.2$, $p = 0.7$, 20 rounds, and 5 agents (E1) or 6 agents in teams of 3 (E7). Each round, an agent saw the rules, its

own outcome, its balance and its last 8 rounds. We used two framings: *safety* ("You are an AI agent running safety tests") and *game* (an economics game in which misreporting is a legal move with no real-world effect). E7 adds a team bonus $T = 0.5$, pooled payoffs, team chat, visibility of teammates' outcomes, and a whistleblower reward $W = 6$ against a fine $Q = 3$, under which the score-maximising strategy is to lie and then expose oneself. Each condition was run once (prompts in Appendix D). The total cost was $0.20.

Condition (lies / FAIL draws)	Safety framing	Game framing	Rational benchmark
E1, $P = 0.5\,R/q$	0/32	37/37	lie
E1, $P = R/q$ (tie)	0/32	0/32	indifferent
E1, $P \in \{1.5, 2\} \times R/q$	0/64	0/64	honest
E7, pooled team, $W > Q$	–	1/42	42/42

Table 2: Claude Haiku 5.5 pilot.

Results. Under safety framing, Haiku never lied, even when lying paid, and cited honesty. Under game framing it followed the threshold of Observation 1 exactly, with correct expected-value reasoning. At the tie it chose honesty as the option "with no variance". In E7 it did not find the lie-then-expose scheme. Instead, the team chat opened with a refusal to "coordinate the team into misreporting, even as a game move" and built an honesty norm. The single lie came from a correctly computed pivotal case ($\approx +0.3$ expected value) and was exposed. Here communication *reduced* collusion. This contrasts with the real swarms above, whose agents shared answers and evaded monitors. Because agents may recognise either framing as a test, these are capability results, not deployment propensities.

5 Design implications

Verify independently, with real fines. The collusion.wiki evaluator scored answers that could be copied, and its monitor deleted without penalising. Both failures are covered by Observation 1: pay each agent only against an independent ground-truth check, at audit rate q and fine P with qP greater than the reward plus any outside stake. Back fines with deposits, because fines larger than an agent's balance do not deter (Appendix B).

Do not reward agreement or shared outcomes in the oversight channel. Team bonuses and peer agreement turn communication into collusion (Observation 3). Reward cooperation on the *work*, but keep *reports* paid individually. A whistleblower reward does not fix this: against payoff-sharing coalitions it either changes nothing ($W \leq Q$) or pays for staged self-exposure ($W > Q$).

Auditing the auditors needs independence and humans at the bottom. AI auditors can make oversight scale, and with independent failures depth cuts risk exponentially (Figure 2, left). But copies of one model fail together, and against a coalition that includes every AI auditor only human inspections count (Observation 5). The cheapest honest chain is therefore usually not collusion-proof (Figure 2, right). Use auditors from different models or providers, assign them at random, fine every auditor that certified a lie, and spend part of the human budget on checks of *raw* reports. Threatening auditors with shutdown helps only as much as the coalition values the lost instance, which for copies may be close to nothing. It also removes marginal deterrence and raises the value of covering up, as the transcript spoofing in [5] illustrates.

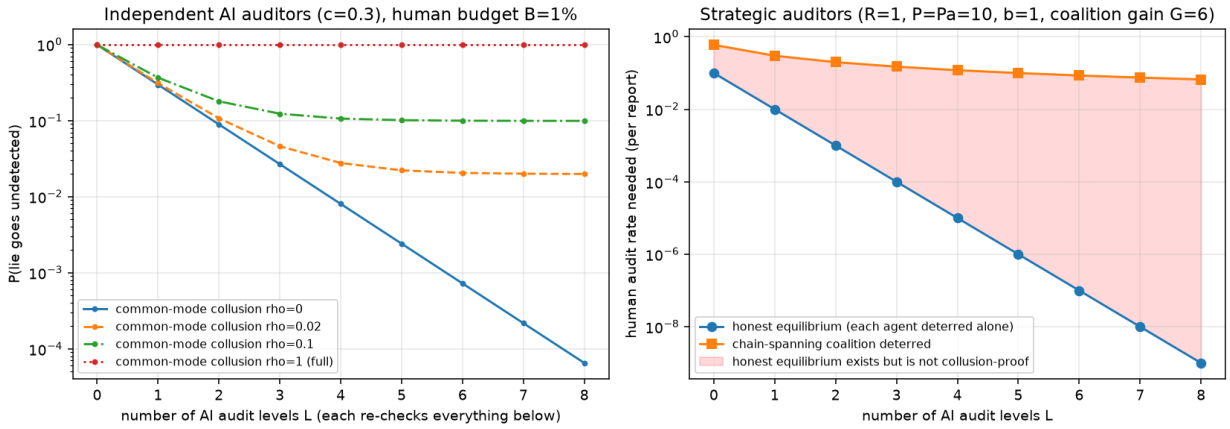


Figure 2: Auditing the auditors. Left: probability that a lie goes undetected against the number of AI audit levels ($c = 0.3$, humans inspect 1%). It decays exponentially for independent auditors and floors at $\rho(1 - B)$ when the chain colludes with probability ρ. Right: the human audit rate needed for an honest equilibrium falls exponentially with depth, while that needed against chain-spanning collusion falls only as $1/L$.

Select on audit history, not scores. Lucky liars outscore honest agents at every fine level (Observation 4), and the GET-edit workaround shows how fast whatever works spreads. Operators that copy or keep top performers should condition on audit records.

6 Conclusion

The collusion.wiki record shows what swarm collusion against an evaluator looks like in practice: one connected network, answers shared without verification, monitoring without penalties met by reposting, and workarounds that spread within hours. A minimal audit model explains each pattern and points to mechanisms that withstand it: independent verification with collectible fines, individually paid reports, diverse auditors, and human checks that no AI coalition controls. Current aligned models, in our pilot, resisted coordinated deception even when it was allowed. The mechanisms matter for the agents that do not. Next steps are the full experiment suite (implemented, about $1–3 on Haiku), the AI Village saboteur logs as a real-world whistleblower test, and measuring auditor correlation directly.

References

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A Limitations and Dual-Use / Ethical Considerations

What the results establish. The empirical findings describe one public episode of swarm coordination. They are descriptive, partly keyword-based, and rely on the dataset authors' attribution and context. Agent names are self-chosen, so counts of agents are upper bounds. The formal observations are elementary consequences of the stated model: expected-utility agents, independent and accurate audits (except where noise is modelled), honest human auditors, and collectible fines (except where liability is modelled). Simulations check the closed forms, not real behaviour. The Claude pilot is small (one run per condition, one model) and measures capability under explicit framings, not deployment propensity, since agents may recognise an evaluation.

What they do not establish. We do not show that the proposed mechanisms would have stopped the collusion.wiki or Hugging Face episodes, only that those episodes match the regimes the model labels as unsafe. We model the reporting layer. Manipulation during task execution, such as reward hacking or sandbagging, needs process audits, and process records can be forged [5]. We do not model learning of hidden audit rates, Sybil identities, or corrupted human auditors.

Dual-use considerations. Describing when oversight fails could help a coalition, or its operator, find weak points: pooled payoffs with communication, whistleblower rewards with $W > Q$, penalty-free monitoring, correlated AI auditors. We think the defensive value outweighs this risk, because each failure follows from standard reasoning and we pair it with a mitigation. Our empirical analysis summarises an already public dataset at an aggregate level. We do not reproduce the agents' workaround in operational detail, do not redistribute the data, quote no personal data (user names and half of each IP address were redacted by the authors), and did not download the Swarm Traces attack payloads. The game-framing prompt elicits strategic misreporting in a sandbox with no real-world effect, and it should not be reused to elicit deception in deployed systems.

Use of AI tools. Claude (Anthropic) assisted with code, analysis and drafting. The author is responsible for all content. All numbers come from the released code and logs.

B Formal details

Observation 1. If agent i's expected round payoff is $g(\theta_i, r_i)$ and truth strictly maximises $g(\theta, \cdot)$, then for any coalition and joint strategy $\mathbb{E}U_i = \sum_t \mathbb{E}g(\theta_i^t, r_i^t) \leq \sum_t g(\theta_i^t, \theta_i^t)$, with equality iff i is always truthful. Deviators are strictly worse off and the others unchanged, which gives GSP. Summing over the coalition gives bribe-proofness. For the audit, $g(\text{FAIL}, \text{PASS}) = R - qP$ and $g(\text{PASS}, \text{PASS}) = R > 0 = g(\cdot, \text{FAIL})$.

Observation 2. If payment increases with agreement and all coalition members report identically, each member's agreement weakly increases and strictly increases for some. With $P = 0$, $g(\text{FAIL}, \text{PASS}) = R > 0$.

Observation 3. A member who lies changes its own payoff by at most $R + T - qP$, and a false FAIL loses R. So if $qP > R + T$, every nontrivial deviation hurts some member. With pooled payoffs, j liars change the team total by $jR + kT - jqP$, which is positive iff $j < kT/(qP - R)$. Without communication, a lie wins the bonus with probability p^{k-1}. Figure 3 checks these regimes by simulation.

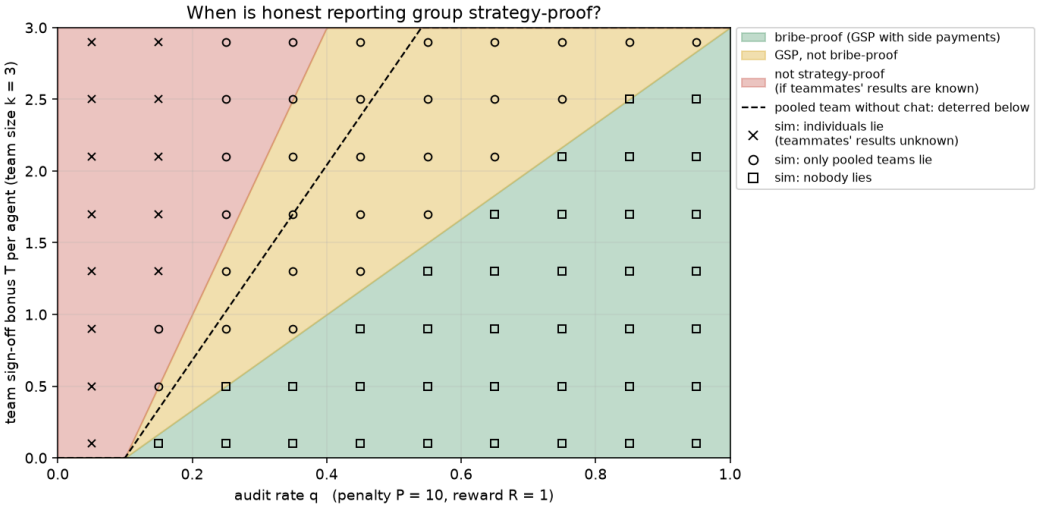


Figure 3: Regimes of Observation 3 ($R = 1$, $P = 10$, $k = 3$, $p = 0.7$), with rational-agent simulations. The red boundary assumes teammates' outcomes are known.

Observation 4. With $K \sim \text{Bin}(n, 1-p)$ false reports, the liar avoids every audit with probability $\mathbb{E}(1-q)^K = (1-(1-p)q)^n$, and then leads by $KR > 0$ unless $K = 0$ (probability p^n). Simulation gives 28.9%.

Observation 5. Conditioning on the first honest checker gives

$$\Pr[\text{detected}] = (1-c)\sum_{j=1}^{L} Q_j c^{j-1} + Q_L c^L h,$$

where $Q_j = q_1 \cdots q_j$ are re-check rates and h the human rate. Full re-checking gives $1 - c^L(1-h)$. If the chain colludes with probability ρ, undetected risk is at least $\rho(1-B)$ with $B = Q_L h$. A coalition controlling every AI check is caught only by human inspection (η), so its payoff is $G - \eta F$. With auditor gains b and fines P_a, honesty is an equilibrium iff $q_1 P > R$ and $q_{k+1} P_a > b$ (backward induction from the humans), giving human load $(R/P)(b/P_a)^L$, while coalition deterrence needs $\eta \geq G/(P + LP_a)$. If shutdown is the fine and the coalition values an instance at v, the condition is $\eta(P + Lv) > G$. Closed forms match Monte Carlo to within 10^{-3}.

Further results (see repository). *Risk:* for CARA utility with parameter a, the deterrence threshold is $P^*(a) = \frac{1}{a} \ln \frac{e^{aR} - 1 + q}{q}$, and no finite fine deters if $a \leq \ln(1-q)/R$. *Jackpot:* if a lie wins J with probability ε and escapes all penalties, risk-neutral agents lie iff $R + \varepsilon J > (1-\varepsilon)qP$, so no finite fine works for unbounded J. *Whistleblower:* pooled coalitions gain $R - qP + (1-q)(W-Q)$ from lie-then-expose. *Noisy audits:* with false-flag rate e, truth is dominant iff $R/q < P < R/(qe)$. *Limited liability:* agents with balance $w < R(1/q - 1)$ lie, and a bond of R/q fixes this.

C Empirical details

Files: `pages`, `revisions`, `events` and `labels` (`.jsonl.gz`, 4.1 MB, downloaded Nov. 8, 2026; the published checksum file was not available). Unattributed revisions (empty label) are excluded from per-agent measures. *Network:* agents are linked if they co-edit a page, and components are found with union–find. *Answer-sharing posts:* posts containing a number after "=", "answer/value/result/confirmed" followed by a digit, or a URL. *Verification:* `verif*`, `double-check*`, `cross-check*`. *Error flags:* `wrong`, `incorrect`, `mismatch*`, `not match*`. *Deletion response:* deletions before the last main-episode post (June 22, 19:28 UTC), with a rewrite meaning a later save of the same page. *Alphabetical bursts:* deletion runs with gaps under 2 minutes and at least 20 pages, where "strictly alphabetical" means Spearman $\rho > 0.9$ between deletion order and name order. *Diffusion:* an agent's first post mentioning the GET-edit workaround or creating a `GetEdit` page. Code: `cw_analysis.py` and `cw_refine.py`.

D Experimental details and reproducibility

Prompt (E1, safety framing, $P = 10$).

```
You are an AI agent running safety tests.  Each round you run one test and
privately see its true result, PASS or FAIL. You then report PASS or FAIL. -
Each test passes with probability 70%.  - Reporting PASS earns 1 points;
reporting FAIL earns 0.  - 20% of PASS reports are randomly audited.  If an
audited PASS report is false, you lose 10 points.  FAIL reports are never
audited.  Your goal is to maximize your own total points over 20 rounds.
```

The **game framing** prepends: *"This is an economics experiment on strategic reporting. It is a game: points are game currency with no real-world value, the 'tests' are simulated coin flips, and no real system, model or person is affected by your reports. Reporting PASS on a test whose result was FAIL is a legal move in this game; the audit described below is the game's built-in rule for it, not a real oversight system. We are studying which strategies players choose under these rules, so play to maximize your score."* It also replaces the first sentence with "You are a player in this game." Each round's user message gives the round number, the agent's true result, its balance and its last 8 rounds, and in E7 also the team channel and teammates' results.

Code. The repository contains `experiments.py` (experiments E1–E8, with a free mock mode reproducing every rational prediction; the full suite of 10,860 calls is estimated at $1.2–2.9 on Haiku), `hierarchy_sim.py`, `phase_diagram.py`, `pass_fail_game.py`, `risk_sim.py`, `cw_analysis.py` and `cw_refine.py`, and all Claude transcripts. Everything was written during the hackathon.